

Asteria Water Resilience Program

Technical Project Report and Synthetic Evaluation Corpus for Retrieval-Augmented Generation

Project code: AWR-2047
Primary contract: CN-AX91 Reporting period: January 2025 - June 2026 Status: Controlled pilot deployment

This fictional report is intentionally designed as a realistic test corpus for document parsing, chunking, semantic retrieval, BM25 keyword search, reranking, context construction, and evidence-grounded question answering. It contains both easy facts and facts distributed across multiple sections.

1. Executive Summary

The Asteria Water Resilience Program (AWR-2047) is a fictional modernization initiative for the North Valley regional water network. The program combines pressure management, leak analytics, telemetry upgrades, and operational decision support. The original approved budget was USD 18.4 million. Following a scope change approved on 14 October 2025, the revised program budget became USD 21.7 million.

The primary delivery contract is CN-AX91. The lead integrator is Meridian Infrastructure Systems, while sensor hardware is supplied by BluePeak Controls. The program director is Dr. Elena Farrow and the technical operations lead is Marcus Hale. The target date for full operational handover is 30 September 2026.

The three highest-ranked program risks are telemetry instability in remote pressure zones, delayed calibration of pressure-reducing valves, and inconsistent asset identifiers across legacy maintenance records. These risks are discussed in detail in Sections 5, 7, and 9. A key design decision was to avoid replacing every legacy sensor; instead, the program prioritizes critical zones and uses analytical validation to identify unreliable signals.

During the pilot, average nightly pressure in Zone N-17 decreased from 53.2 mH₂O to 46.8 mH₂O without violating the minimum service threshold of 32.0 mH₂O. The observed minimum during the controlled trial was 34.6 mH₂O. Estimated background leakage in that zone declined by 11.8 percent over the twelve-week comparison period.

1.1 Key Outcomes

- AWR-2047 remained within the revised USD 21.7 million budget as of 30 June 2026.
- Contract CN-AX91 is the principal systems-integration agreement.
- Pilot Zone N-17 achieved an 11.8 percent estimated reduction in background leakage.
- Full operational handover is planned for 30 September 2026.
- The most consequential unresolved risk is intermittent telemetry loss at gateway GW-17B.

2. Program Background and Objectives

North Valley operates a mixed-age distribution network that includes assets commissioned between 1978 and 2024. Pressure variation, inconsistent telemetry, and incomplete maintenance metadata made it difficult for operations staff to compare hydraulic behavior across zones. AWR-2047 was initiated to improve operational visibility without requiring a complete replacement of field infrastructure.

The program has four formal objectives. First, reduce avoidable pressure-related leakage while preserving customer service levels. Second, improve confidence in telemetry used for operational decisions. Third, create a repeatable workflow for prioritizing field interventions. Fourth, provide evidence-based reporting that links analytical recommendations to source measurements, work orders, and engineering approvals.

The approved design principle is 'measure, verify, then automate.' Automated recommendations may be generated by analytical systems, but pressure setpoint changes require validation against hydraulic constraints and operational policy. The program does not permit autonomous field actuation during the pilot.

2.1 Scope Boundaries

The scope includes 14 pressure management zones, 63 pressure sensors, 18 flow meters, 12 pressure-reducing valves, two telemetry gateways, and a central analytics environment. Customer billing systems, treatment plant control logic, and chemical dosing systems are explicitly outside the program scope.

2.2 Success Criteria

Metric	Target	Pilot Status
Night pressure reduction	At least 4.0 mH2O in selected zones	6.4 mH2O in N-17
Minimum service pressure	Never below 32.0 mH2O	Minimum 34.6 mH2O
Telemetry completeness	$\geq 98.5\%$	97.9% overall; 94.8% at GW-17B path
Leakage reduction	$\geq 8\%$ estimated in pilot zone	11.8% estimated
Critical incident response	< 15 minutes	Median 9.6 minutes

3. Technical Architecture

The solution architecture separates field acquisition, data validation, analytical services, and human approval. Sensors transmit observations through encrypted gateway channels to the ingestion layer. The ingestion service validates timestamps, range limits, and asset identifiers before data enters the analytical store.

Pressure and flow observations are retained at one-minute resolution for 45 days and then aggregated to five-minute resolution for long-term analysis. Maintenance work orders are joined using canonical asset identifiers. A mapping table is maintained because legacy systems use at least three identifier formats for the same valve family.

3.1 Data Processing Stages

- Acquisition: collect pressure, flow, valve-state, and gateway-health signals.
- Validation: reject impossible values, flag stale timestamps, and identify sensor dropouts.
- Normalization: map asset identifiers to the canonical registry.
- Analytics: calculate pressure trends, leakage indicators, and confidence scores.
- Decision support: present recommendations with evidence and uncertainty.

AWR-2047 intentionally uses a hybrid rule-and-model approach. Deterministic engineering constraints define non-negotiable limits, while statistical and machine-learning components rank opportunities. This approach was selected because the pilot team required traceable recommendations rather than opaque optimization.

3.2 Critical Identifiers

Identifier	Meaning	Notes
AWR-2047	Program code	Primary project identifier
CN-AX91	Systems integration contract	Meridian Infrastructure Systems
N-17	Pilot pressure zone	Primary pressure-management trial
PRV-17-03	Primary pressure-reducing valve	Pilot control asset
GW-17B	Remote telemetry gateway	Known intermittent packet-loss issue
INC-4472	Telemetry incident	Occurred 18 February 2026

4. Schedule, Budget, and Procurement

The baseline program schedule was approved on 8 January 2025. Field deployment began on 3 March 2025, and the original handover target was 30 June 2026. Two factors caused schedule revision: delayed radio licensing for the western telemetry corridor and additional commissioning work required for PRV-17-03. The steering committee approved a revised operational handover date of 30 September 2026.

The original budget of USD 18.4 million covered field hardware, integration, analytics, testing, training, and contingency. On 14 October 2025, change request CR-28 added redundant telemetry paths for three remote zones and expanded cybersecurity testing. The approved revised budget became USD 21.7 million. The increase was USD 3.3 million.

4.1 Budget Breakdown

Category	Original (USD M)	Revised (USD M)	Reason for Change
Field instrumentation	5.1	5.8	Additional gateway redundancy
Systems integration	4.8	5.2	Extended commissioning
Analytics platform	2.7	3.0	Additional validation tooling
Cybersecurity	1.4	2.1	Expanded penetration and resilience testing
Training and change	1.2	1.3	Additional operator workshops
Contingency	3.2	4.3	Schedule and supply-chain exposure

4.2 Procurement Notes

Meridian Infrastructure Systems is the lead integrator under CN-AX91. BluePeak Controls supplies pressure and flow instrumentation under purchase order PO-BP-552. A separate cybersecurity review was awarded to Aegis Harbor Labs under engagement AH-77. The contractual service-level target for critical software incidents is a response within 15 minutes.

5. Risk Register

The risk register is reviewed every two weeks. Risks are scored using probability and impact on a five-point scale, with the product used as the priority score. The steering committee treats scores of 15 or above as critical. As of 30 June 2026, two risks remain critical and four remain high.

Risk ID	Description	Prob.	Impact	Score	Mitigation
R-07	Intermittent telemetry loss through GW-17B	4	5	20	Redundant path + buffer replay
R-12	PRV calibration drift after maintenance	4	4	16	Monthly verification
R-19	Legacy asset-ID mismatch	3	4	12	Canonical mapping registry
R-23	Operator over-reliance on model score	3	4	12	Human approval + confidence display
R-31	Vendor firmware delay	2	4	8	Approved fallback release

Risk R-07 is the highest-priority operational risk. Packet loss is concentrated on the GW-17B route during periods of high radio interference. The mitigation design stores up to 90 minutes of observations at the edge and replays missing data after connectivity returns. However, replay does not support real-time decision making, so the project added a redundant communications path under change request CR-28.

Risk R-12 concerns calibration drift on PRV-17-03 after maintenance. The issue does not indicate uncontrolled actuation; rather, the reported valve position can differ from the mechanically verified position by up to 4.2 percentage points. The current control measure is monthly manual verification and recalibration after any actuator service.

Risk R-23 is a human-factors risk. During an April 2026 workshop, two operators treated a low-confidence anomaly score as if it were a confirmed leak. The user interface was changed to display confidence bands and require an explicit acknowledgement when evidence quality is below the operational threshold.

6. Pilot Zone N-17 Results

Zone N-17 was selected because it has stable customer demand patterns, a controllable PRV, and sufficient historical measurements for comparison. The primary control asset is PRV-17-03. The pilot evaluation period covers twelve weeks before and twelve weeks after the revised nighttime setpoint strategy.

6.1 Pressure Performance

Indicator	Before	After	Change
Average nightly pressure	53.2 mH2O	46.8 mH2O	-6.4 mH2O
95th percentile pressure	58.7 mH2O	50.9 mH2O	-7.8 mH2O
Minimum observed pressure	36.1 mH2O	34.6 mH2O	-1.5 mH2O
Service threshold	32.0 mH2O	32.0 mH2O	No change

No validated service-pressure violation occurred during the trial. Three raw observations appeared below 32.0 mH2O, but each was associated with sensor reset events and was excluded after cross-checking neighboring sensors and gateway logs.

6.2 Leakage Indicator

The background-leakage estimate decreased by 11.8 percent after pressure optimization. This figure is an engineering estimate derived from minimum-night-flow normalization and should not be interpreted as a direct metered volume saving. The confidence interval is wider during weeks with unusual industrial demand.

The largest single-week reduction was 15.3 percent in week 9. The smallest was 6.7 percent in week 3. The program team considers the 11.8 percent twelve-week estimate sufficient to pass the pilot success criterion of at least 8 percent.

7. Telemetry Reliability and Incident INC-4472

On 18 February 2026, incident INC-4472 caused intermittent loss of observations from six devices connected through gateway GW-17B. The first missing packet was detected at 02:14 local time. Automated monitoring created an alert at 02:17, and the on-call engineer acknowledged it at 02:22.

The immediate cause was radio interference combined with a firmware retry defect. The gateway itself remained powered and retained local data. Connectivity returned at 03:06, after which the edge buffer replayed 97.4 percent of the missing observations. The remaining records could not be reconstructed.

The incident did not cause an unsafe pressure change because the pilot architecture prohibits autonomous field actuation when telemetry confidence is degraded. PRV-17-03 remained on its previously approved setpoint until operators confirmed stable data.

7.1 Corrective Actions

- Upgrade gateway firmware from 3.8.1 to validated release 3.8.4.
- Deploy a redundant telemetry path for GW-17B.
- Block optimization recommendations when data completeness over the preceding 20 minutes is below 97 percent.
- Retain edge buffers for at least 90 minutes.

The firmware upgrade was completed on 11 March 2026. The redundant path was commissioned on 26 May 2026. Since commissioning, no incident with more than eight consecutive minutes of data loss has been recorded on the GW-17B route.

8. Data Quality and Asset Identity

Historical maintenance systems used inconsistent asset identifiers. For example, PRV-17-03 appears as 'PRV1703', 'Valve 17-3', and 'N17-PRV03' in older records. These aliases created incorrect joins during early analytics tests and caused maintenance events to be associated with the wrong asset.

The project introduced a canonical asset registry. Each physical asset receives one stable identifier and a list of known aliases. Incoming work orders and telemetry records are normalized against this registry before analytical processing. Records that cannot be resolved with sufficient confidence are quarantined for review.

8.1 Data Quality Thresholds

Check	Threshold	Action if Failed
Timestamp freshness	<= 3 minutes	Mark observation stale
Pressure range	0 - 120 mH2O	Reject + alert
Flow range	Site-specific bounds	Flag for review
20-minute completeness	>= 97%	Allow optimization recommendation
Asset-ID resolution confidence	>= 0.92	Auto-map; otherwise quarantine

Overall telemetry completeness for the reporting period was 97.9 percent. The target is 98.5 percent, so the program has not yet achieved the formal telemetry-completeness objective. The primary contributor is the historical performance of GW-17B. Excluding that path, completeness was 99.1 percent.

9. Cybersecurity and Access Control

The cybersecurity design follows least-privilege access and separates operational technology from analytics services. Field gateways initiate outbound encrypted connections; inbound administrative access to gateways is disabled during normal operations. Privileged maintenance requires a time-limited approval token and is logged.

Aegis Harbor Labs performed a controlled assessment under engagement AH-77 in April 2026. No critical remote-code-execution finding was identified. Two high-severity findings were reported: an overly permissive service account in the staging environment and insufficient rotation of a non-production API credential.

The service-account finding was closed on 29 April 2026. The credential-rotation control was closed on 6 May 2026. The final reassessment concluded that no unresolved high-severity findings remained, although three medium-severity hardening recommendations were carried into the production-readiness backlog.

9.1 Data Protection

- All external service connections use TLS 1.3 where supported.
- Operational data at rest is encrypted using managed keys.
- User access is role-based and scoped by operational region.
- Administrative events and evidence-view actions are retained in the audit log for 365 days.

10. Analytical Decision Support

The analytical service does not directly command field devices. It generates ranked recommendations accompanied by source measurements, confidence indicators, and constraint checks. Operators review these recommendations before any approved change is sent through the operational workflow.

A recommendation is suppressed when any hard safety constraint fails. Examples include inadequate telemetry completeness, predicted service pressure below 32.0 mH₂O, unresolved sensor conflict, or an active maintenance lock on the target valve.

10.1 Recommendation Confidence

Confidence is calculated from telemetry completeness, sensor agreement, model stability, and historical similarity. A score above 0.85 is considered high confidence, 0.70 to 0.85 is moderate, and below 0.70 is low. Low-confidence results may still be displayed for investigation but are clearly labeled and cannot be approved without additional evidence.

10.2 Example Recommendation

On 4 June 2026, the system recommended reducing the nighttime setpoint of PRV-17-03 by 1.5 mH₂O. The recommendation confidence was 0.89. Hydraulic checks predicted a minimum pressure of 35.2 mH₂O, above the 32.0 mH₂O policy threshold. An operator approved the recommendation after comparing the evidence with the previous seven nights.

11. Operations, Governance, and Human Oversight

The operations model assigns clear responsibilities. The program director, Dr. Elena Farrow, approves major scope changes and risk acceptance. Marcus Hale, technical operations lead, owns commissioning readiness and incident coordination. Data-quality exceptions are reviewed by the analytics engineering team.

The steering committee meets monthly, while the technical risk group meets every two weeks. Any change that modifies a safety threshold, automated recommendation policy, or field-control boundary requires documented engineering approval.

11.1 Escalation Rules

- Critical telemetry failure: notify technical operations within 5 minutes.
- Potential service-pressure violation: suspend recommendation generation for the affected zone.
- Unresolved data-identity conflict: quarantine affected records.
- Cybersecurity finding rated high or critical: block production release until formally closed or risk-accepted.

The program maintains a decision log linking major recommendations to evidence and approvals. This requirement was introduced after an internal review found that technically correct decisions were difficult to reconstruct when rationale was recorded only in email.

12. Lessons Learned and Remaining Work

The pilot demonstrated that pressure optimization can create measurable benefits without replacing all field assets. However, analytical quality depends strongly on telemetry reliability and asset identity. The project team concluded that data quality must be treated as an operational capability rather than a one-time cleanup task.

A second lesson is that user trust improves when recommendations expose evidence. Operators responded more positively to recommendations that included the relevant pressure trend, confidence score, and constraint check. A single unexplained model score was considered insufficient for operational approval.

The principal remaining tasks before handover are completion of training in the western zones, closure of the three medium-severity cybersecurity recommendations, final acceptance testing of redundant telemetry, and confirmation that telemetry completeness reaches or exceeds the 98.5 percent objective for four consecutive weeks.

12.1 Handover Readiness

Item	Status	Target
N-17 pressure pilot	Complete	Completed 30 May 2026
GW-17B redundant path	Complete	Commissioned 26 May 2026
Operator training	In progress	31 August 2026
Telemetry objective	Not yet met	4 consecutive weeks $\geq$ 98.5%
Operational handover	Planned	30 September 2026

Appendix A. Reference Facts and Suggested Test Questions

This appendix contains concise reference information intended to support deterministic retrieval tests. The same facts also appear elsewhere in the report, sometimes with different wording.

Question Topic	Reference Fact
Project code	AWR-2047
Primary contract	CN-AX91
Program director	Dr. Elena Farrow
Technical operations lead	Marcus Hale
Original budget	USD 18.4 million
Revised budget	USD 21.7 million
Budget approval date	14 October 2025
Pilot zone	N-17
Primary PRV	PRV-17-03
Problem gateway	GW-17B
Incident	INC-4472 on 18 February 2026
Night pressure before	53.2 mH2O
Night pressure after	46.8 mH2O
Minimum service threshold	32.0 mH2O
Leakage estimate reduction	11.8%
Telemetry target	98.5%
Observed telemetry completeness	97.9%
Handover date	30 September 2026

Suggested Local Retrieval Questions

- What is the project code?
- What is contract CN-AX91?
- Who is the technical operations lead?
- What happened during incident INC-4472?
- What is the minimum service-pressure threshold?
- How much did average nightly pressure change in Zone N-17?

Suggested Multi-Evidence Questions

- Why was the budget increased, and what was the final revised amount?
- Which issues still prevent the project from being considered fully ready for handover?
- How did telemetry reliability affect the pressure-optimization process?
- What controls prevent an analytical recommendation from directly causing an unsafe field action?